

JEE MAIN REVISION MASTER: HYDROCARBONS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 166 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Hydrocarbons - Core Application	166 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Hydrocarbons - Core Application (Very High Yield — 166 PYQs)

Core Principle: Key problem pattern in Hydrocarbons.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The major product of which of the following reaction is not obtained by rearrangement reaction?..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Identify the correct IUPAC name of hydrocarbon (x) containing three primary carbon atoms and with molar mass 72gmol^{-1} " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Arrange the following alkenes in decreasing order of stability. Choose the correct answer from the options given below:..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 23rd January Evening Shift (Q#1)

Consider the following reaction of benzene. In compound (Q), the percentage of oxygen is ____ %. (Nearest integer)

Correct Answer: 10

Solution Analysis: Molecular mass of compound Q is 157. In Q, oxygen contributes mass 16 (for one oxygen atom). So, percentage of oxygen in Q is %of oxygen in product 'Q' is $= (16 / 157) \times 100 = 10.19\%$ Nearest integer = 10 %.